Complaint.java

Java

```java
package com.example.complaint_management;

public class Complaint {
    private int id;
    private String description;
    private String category;
    private String status;

    public Complaint() {
    }

    public Complaint(int id, String description, String category, String status) {
        this.id = id;
        this.description = description;
        this.category = category;
        this.status = status;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getDescription() {
        return description;
    }

    public void setDescription(String description) {
        this.description = description;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

    public String getStatus() {
        return status;
    }

    public void setStatus(String status) {
        this.status = status;
    }

}

```

ComplaintController.java

Java

```java
package com.example.complaint_management;

import java.util.ArrayList;
import org.springframework.http.HttpStatus;
import org.springframework.web.bind.annotation.*;
import org.springframework.web.server.ResponseStatusException;

@RestController
@RequestMapping("/complaints")
public class ComplaintController {
    private ArrayList<Complaint> complaints = new ArrayList<>();
    private int nextId = 1;

    @PostMapping
    public Complaint registerComplaint(@RequestBody Complaint complaint) {

        if (complaint.getDescription() == null ||
            complaint.getDescription().trim().isEmpty()) {
            throw new ResponseStatusException(
                HttpStatus.BAD_REQUEST, "Description is required");
        }

        if (complaint.getCategory() == null ||
            complaint.getCategory().trim().isEmpty()) {
            throw new ResponseStatusException(
                HttpStatus.BAD_REQUEST, "Category is required");
        }

        String category = complaint.getCategory().trim();

        if (!category.equalsIgnoreCase("Classroom") &&
            !category.equalsIgnoreCase("Wi-Fi") &&
            !category.equalsIgnoreCase("Library")) {
            throw new ResponseStatusException(
                HttpStatus.BAD_REQUEST,
                "Category must be Classroom, Wi-Fi, or Library");
        }

        Complaint newComplaint = new Complaint(
            nextId++,
            complaint.getDescription().trim(),
            category,
            "PENDING"
        );

        complaints.add(newComplaint);
        return newComplaint;
    }

    @GetMapping
    public ArrayList<Complaint> getAllComplaints() {
        return complaints;
```

ComplaintController.java

Java

```
53          }
54
55      }
56
```

ComplaintManagementApplication.java

Java

```java
package com.example.complaint_management;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class ComplaintManagementApplication {

    public static void main(String[] args) {
        SpringApplication.run(ComplaintManagementApplication.class, args);
    }

}

```